

The Derivative from First Principles

ANSWER KEY



The definition of the derivative

- 1 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 + 3x$.
$$\frac{(x+h)^2 + 3(x+h) - (x^2 + 3x)}{h} = \frac{2xh + h^2 + 3h}{h} = 2x + h + 3 \rightarrow 2x + 3.$$
- 2 Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x}$ (for $x > 0$).
$$\frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{h}{h(\sqrt{x+h} + \sqrt{x})} = \frac{1}{\sqrt{x+h} + \sqrt{x}} \rightarrow \frac{1}{2\sqrt{x}}.$$
- 3 The graph shows $f(x) = x^2$ with the secant line through $(1, 1)$ and $(3, 9)$.
 - a Find the slope of the secant line. $\frac{9-1}{3-1} = 4.$
 - b Find $f'(2)$ using the derivative, and compare it to the secant slope. $f'(x) = 2x$, so $f'(2) = 4$ — for this quadratic, the secant slope between $x = 1$ and $x = 3$ happens to equal the instantaneous slope at the midpoint $x = 2$.

Differentiability

- 4 Explain why $f(x) = |x|$ is not differentiable at $x = 0$, using one-sided derivatives.
The left-hand derivative is $\lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1$ and the right-hand derivative is $\lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1$. Since these disagree, $f'(0)$ does not exist.
- 5 State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?
Differentiability implies continuity (a corner or asymptote would break the limit needed for the derivative).
The converse is false — $f(x) = |x|$ is continuous at 0 but not differentiable there.

More derivatives from the definition

- 6 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x}$ (for $x \neq 0$).
$$\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{x - (x+h)}{hx(x+h)} = \frac{-h}{hx(x+h)} = \frac{-1}{x(x+h)} \rightarrow -\frac{1}{x^2}.$$
- 7 Use the limit definition to find $f'(x)$ for $f(x) = x^3$.
$$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3, \text{ so } \frac{f(x+h) - f(x)}{h} = \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2 \rightarrow 3x^2.$$

Interpreting the derivative

- 8 A ball's height is $h(t) = -5t^2 + 20t + 2$ (metres, seconds). Use the derivative definition idea to explain what $h'(2)$ represents physically, and find its value using $h'(t) = -10t + 20$.
- $h'(2)$ represents the instantaneous vertical velocity of the ball at $t = 2$ seconds.
- $h'(2) = -10(2) + 20 = 0$ m/s — the ball is momentarily at rest, at its peak height.